metals

Purpose

The heavy metals module simulates user-listed metals across HRU soils and plants. The set of metals simulated is declared in constituents.cs, which populates `cs_db%num_metals`. Only initial soil and plant concentrations are loaded by a dedicated reader; no separate metal parameter database is read by `hmet_hru_aqu_read`.

Activation

`in_parmdb%hmetcom_db` is defined in src/input_file_module.f90 with default `metals.mtl`. No reader in `src/` opens this filename, so the parameter database is declared in `file.cio` but not loaded.

The initial-conditions file is referenced from `file.cio` via `in_init%hmet_soil` (default `hmet_hru.ini`). `hmet_hru_aqu_read` is called from `proc_read.f90`.

Files

hmet_hru.ini

Initial metal concentrations for HRU soils and plants.

- Reader: src/hmet_hru_aqu_read.f90
- Type: `hmet_soil_ini` (`type cs_soil_init_concentrations`) in src/constituent_mass_module.f90. Fields: `name`, `soil(num_metals)` (ppm), `plt(num_metals)` (ppm).
- Format: one title line, then per record: a header line, a name line, then `num_metals` pairs of lines (`metal_name, soil_value` then `metal_name, plt_value`) in `constituents.cs` order.

The same reader allocates `cs_hmet_solsor(num_metals)`.

metals.mtl

Declared as `in_parmdb%hmetcom_db` in `input_file_module.f90` (default name `metals.mtl`). No reader in the current source opens this file. The file is reserved but inert.

Related

- constituents.cs. Declares the metal list and sets `cs_db%num_metals`. The metals block in `constit_db_read` is read but its name-database cross-walk is commented out, so listed metals only contribute the count.
- pesticide module. Same per-object initial-condition pattern.
- pathogen module. Same per-object initial-condition pattern.